

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

### GROUP 'A' [11 × 1 = 11] · Time: 25 min

Choose the best alternative.

- In  $F = a\sqrt{x} + bt^2$ , where  $F$  is force,  $x$  is distance and  $t$  is time, the dimensions of  $b/a$  are  
(a)  $[L^{1/2} T^{-2}]$  (b)  $[M L^{1/2} T^{-2}]$  (c)  $[L^{-1/2} T^2]$  (d)  $[M L T^{-4}]$
- Two forces of 6 N and 8 N act at a point. Which of these cannot be their resultant?  
(a) 2 N (b) 10 N (c) 14 N (d) 15 N
- The angle between  $A = \hat{i} + \hat{j}$  and  $B = \hat{i} - \hat{j}$  is  
(a)  $0^\circ$  (b)  $45^\circ$  (c)  $90^\circ$  (d)  $180^\circ$
- A car accelerates uniformly from rest to  $20 \text{ m s}^{-1}$  and then moves uniformly for 5 s. The distance covered in 10 s is  
(a) 100 m (b) 150 m (c) 200 m (d) 250 m
- At the highest point of an oblique projectile, the angle between velocity and acceleration is  
(a)  $0^\circ$  (b)  $45^\circ$  (c)  $90^\circ$  (d)  $180^\circ$
- A ball is projected at  $45^\circ$  with  $20 \text{ m s}^{-1}$  ( $g = 10 \text{ m s}^{-2}$ ). Its maximum height is  
(a) 5 m (b) 10 m (c) 20 m (d) 40 m
- On scale X, ice point is  $20^\circ\text{X}$  and steam point is  $220^\circ\text{X}$ .  $50^\circ\text{C}$  on this scale reads  
(a)  $100^\circ\text{X}$  (b)  $110^\circ\text{X}$  (c)  $120^\circ\text{X}$  (d)  $140^\circ\text{X}$
- 100 g of water at  $80^\circ\text{C}$  is mixed with 300 g of water at  $20^\circ\text{C}$ . The final temperature is  
(a)  $35^\circ\text{C}$  (b)  $40^\circ\text{C}$  (c)  $50^\circ\text{C}$  (d)  $60^\circ\text{C}$
- An object is placed 40 cm in front of a concave mirror of radius of curvature 40 cm. The image is  
(a) virtual, erect, magnified (b) real, inverted, same size (c) real, inverted, diminished (d) at infinity
- A charge  $q$  is placed at the centre of a cube. The flux through one face is  
(a)  $q/\epsilon_0$  (b)  $q/6\epsilon_0$  (c)  $6q/\epsilon_0$  (d) zero
- A body acquires a charge of  $-3.2 \text{ }\mu\text{C}$ . The number of electrons it gained is

$$2i \quad i+i-j \quad 2$$

$$2i+0j$$

MCQ

Q

$$1) \rightarrow a) L^{\frac{1}{2}} T^{-2}$$

$$2) \rightarrow a) 15$$

$$3) \rightarrow a) 0^\circ$$

$$4) \rightarrow b) 150m$$

$$\rightarrow \textcircled{5) a) 0^\circ}$$

$$\rightarrow b) 10m$$

$$\rightarrow c) 120^\circ X$$

$$\rightarrow a) 35^\circ C$$

$$\rightarrow b) \text{ real, inverted same size}$$

$$10) \rightarrow b) 9/630 \quad \rightarrow \text{II) } a) 2 \times 10^{13}$$

## Mock

### Q-3

12)

a)

→

The principle of homogeneity states that, the dimensions of the physical quantities in the both sides of the physical equation must be equal.

Eg:  $F = ma$

$$[F] = [m][a]$$

→ To check correctness of  $T = 2\pi \sqrt{\frac{L}{g}}$

$$[T] = [M^0 L^0 T]$$

$$[2\pi \sqrt{\frac{L}{g}}] = [2\pi] \sqrt{\frac{[L]}{[g]}}$$

$$= \sqrt{\frac{[L]}{[L T^{-2}]}}$$

$$= \sqrt{[T^2]}$$

$$= [T] = [M^0 L^0 T]$$

∴ Correct

$T = \frac{\text{Stress}}{\text{Strain}}$

$$\frac{d}{dt} \left( \frac{1}{2} m \dot{x}^2 \right) = \frac{d}{dt} \left( \frac{1}{2} m \dot{x}^2 \right)$$

$$f \propto g_a^{ab} T_{bc}$$

$$[T^{-1} \log U] = [T^{-1}] = [T]^{-1} = [T]$$

$$\begin{aligned} \text{comp } [KJTu] &= [LJ][LT][Ly]^b \\ &= [LJ][ML^{-1}T^{-2}][M L^{-1}]^b \\ &= [M]^{b+c}[L]^{a+b-c}[T]^c \end{aligned}$$

$$\begin{aligned} b+c &= 0, & a+b-c &= 0-2b = -1 \\ c &= -\frac{1}{2}, & a &= 1 \\ b &= \frac{1}{2} \end{aligned}$$

$$\frac{a+1}{2} + 1 = 0$$

$$\frac{1}{2} - 1 = -\frac{1}{2} \quad \frac{1}{2} - 1 = -\frac{1}{2}$$

$$f = k \cdot T^{\frac{1}{2}} \cdot e^{\frac{-E_a}{2RT}}$$

$$f = \frac{K}{2\sqrt{y}}$$

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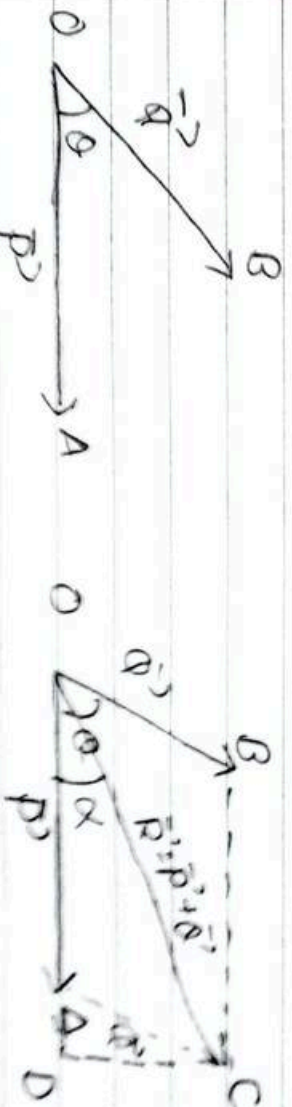
a1

→ Parallelogram law of vector addition states that, when two vector act at upon a particle at an angle  $\theta$ , representing its magnitude and direction by the two adjacent sides of a parallelogram, then the diagonal of the parallelogram from that point gives the resultant vector."

Let, P and Q be vectors acting at an angle  $\theta$  be represented by its magnitude and direction by the two adjacent sides OA and OB of the parallelogram OACD, then, OC gives the resultant R.

$$\vec{R} = \vec{P} + \vec{Q}$$

$$OC^2 = OA^2 + OB^2$$



Now, let us draw, CD  $\perp$  OA extended to D, then, from right angle triangle OCD

$$OC^2 = OD^2 + CD^2$$

$$= (OA + AD)^2 + DC^2$$

Again from,  $\triangle ADC$ ,

$$\sin \theta = \frac{CD}{AC}$$

$$\vec{CD} = AC \sin \theta \quad (ii)$$

$$\cos \theta = \frac{AD}{AC}$$

$$\vec{AD} = AC \cos \theta \quad (iii)$$

Now Putting (ii) and (iii) in eq (i)

$$\begin{aligned} N_c \quad R^2 &= (\vec{P} + \vec{AD} \cos \theta)^2 + AC^2 \sin^2 \theta \\ &= P^2 + 2\vec{P} \cdot \vec{AD} \cos \theta + \vec{AD}^2 \cos^2 \theta + AC^2 \sin^2 \theta \\ &= P^2 + 2PQ \cos \theta + Q^2 (\sin^2 \theta + \cos^2 \theta) \\ |\vec{R}| &= \sqrt{P^2 + 2PQ \cos \theta + Q^2} \end{aligned}$$

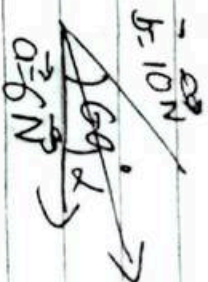
For direction, In  $\triangle ODC$

$$\phi \text{ and } = \frac{OD}{CD}$$

$$= \frac{AC \sin \theta}{OP + AD} = \frac{Q \sin \theta}{P + Q \cos \theta}$$

b1  
→ Given,

$$\vec{a} = 6\text{N}$$
$$\vec{b} = 10\text{N}$$



$$R = \sqrt{a^2 + b^2 + 2\vec{a}\vec{b}\cos 60}$$
$$= \sqrt{6^2 + 10^2 + 2 \times 6 \times 10 \cos 60}$$
$$= 14\text{N}$$

$$\text{Dir, } \alpha = \tan^{-1} \left( \frac{a \sin \theta}{b + a \cos \theta} \right)$$
$$= \tan^{-1} \left( \frac{6 \sin 60}{10 + 6 \cos 60} \right)$$
$$= 21.7867^\circ \text{ from } \vec{a}$$

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→ a1

Resolution of a vector is the process of splitting a vector into two or more vectors, which if combined gives the original vector.

→ Resolving a vector into two rectangular components.

Let,  $\vec{P}$  be the vector, and a rectangle  $OABC$  is formed joining the line,  $OC(\vec{P})$  such that  $OB \perp OA$ ,  $\theta = 90^\circ$



Now, let the  $OA$  be  $\vec{x}$  and  $OB$  be  $\vec{y}$  then, from right angled triangle,  $OAC$ ,

$$\sin \theta = \frac{AC}{OC} \quad \text{since, } AC = OB$$

$$\sin \theta = \frac{OB}{OC}$$

$$OC \sin \theta = OB$$

$$|\vec{P}| \sin \theta = |\vec{y}|$$

$$y = p \sin \theta, \quad \text{--- (i)}$$

$$\cos \theta = \frac{OA}{OC}$$

$$\cos \theta = \frac{|\vec{x}|}{|\vec{P}|}$$

$$p \cos \theta = x$$

$$x = p \cos \theta \quad \text{--- (ii)}$$

From (i) and (ii)  
vertical component =  $p \sin \theta$   
horizontal component =  $p \cos \theta$

Ans.

$$\frac{A}{B} = \frac{2}{3} \quad \sqrt{2^2 + 3^2}$$

b1  
→ Soln, Given,

$$A = 2\hat{i} + 3\hat{j} - \hat{k}$$

$$B = \hat{i} - 2\hat{j} + 4\hat{k}$$

$$\theta = ?$$

We know,

$$A \cdot B = AB \cos \theta$$

$$\sin \theta = \frac{|A \times B|}{|A||B|} \quad \cos \theta = \frac{A \cdot B}{|A||B|}$$

$$\theta = \sin^{-1} \left( \frac{|A \times B|}{|A||B|} \right)$$

$$|A \times B| = |(2\hat{i} + 3\hat{j} - \hat{k}) \times (\hat{i} - 2\hat{j} + 4\hat{k})|$$

$$= |(2\hat{i} - 6\hat{j} - 4\hat{k})|$$

$$= \sqrt{(-8)^2 + (-6)^2 + (-4)^2}$$

$$AB = \sqrt{(2^2 + 3^2 + 1^2)} \sqrt{(1^2 + 2^2 + 4^2)}$$

$$= \sqrt{(4 + 9 + 1)} \sqrt{(1 + 4 + 16)}$$

$$= \sqrt{14} \sqrt{21}$$

$$= 17.32$$

$$\theta = \sin^{-1} \left( \frac{-8}{17.32} \right) \quad \theta = \cos^{-1} \left( \frac{-8}{17.32} \right)$$

$$= 117.81184^\circ$$

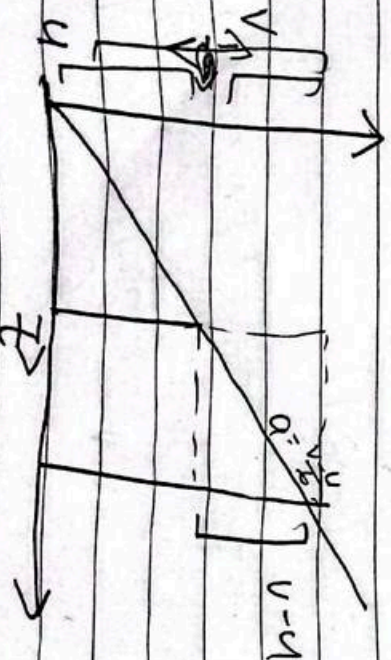
c1  
->

No a rectangular component ~~ratio~~ of a vector cannot be greater than the vector itself either it can be smaller or same, because for eq, horizontal component =  $|\vec{a}| \cos \theta$ , here,  $|\vec{a}| \cos \theta \leq |\vec{a}|$ ,

15)

a1 ~~th~~ <sup>th</sup> ~~th~~ <sup>th</sup>

-> Soln, a velocity time graph is drawn,



We know,  $s = \int u \, dt$   ~~$s = \int u \, dt$~~   $s = \int u \, dt$

So the area under the graph gives

the distance,

$$s = \frac{1}{2}(v-u)t$$

$$= \frac{1}{2}(v-u)\left(\frac{v-u}{a}\right)$$

$$= \frac{1}{2}(v-u) \cdot t$$

$$= \frac{1}{2}(v-u) \cdot \frac{v-u}{a}$$

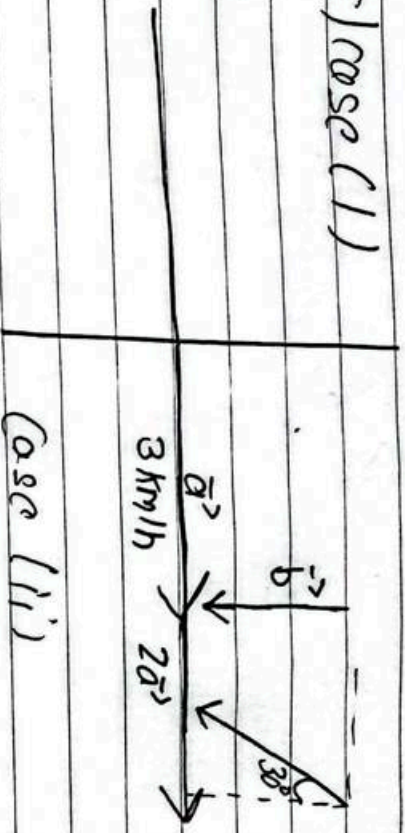
b)

From eq (1)

$$\vec{a} = 3\hat{i}$$

$$\vec{b} = ?$$

$$\theta = 90^\circ$$



Case (ii)

$$a, \vec{a} = 3\hat{i}$$

$$\vec{b} = -4\hat{j}$$

$$\vec{a} + \vec{b} = 3\hat{i} - 4\hat{j}$$

$$\vec{a} = 6\hat{i}$$

$$\vec{b} = -b \sin 30^\circ \hat{i} + b \cos 30^\circ \hat{j}$$

$$\vec{a} + \vec{b}$$

$$= 6\hat{i} + b \sin 30^\circ \hat{i} + b \cos 30^\circ \hat{j}$$

$$= (6 + b \sin 30^\circ) \hat{i} + b \cos 30^\circ \hat{j}$$

$$3\hat{i} - 4\hat{j} = (6 + b \sin 30^\circ) \hat{i} + b \cos 30^\circ \hat{j}$$

$$R = \sqrt{3^2 + 4^2}$$

$$R = \sqrt{(6 - b \cos 30^\circ)^2 + (b \sin 30^\circ)^2}$$

$$R = 1$$

$$25 = (6 - b \cos 30^\circ)^2 + (b \sin 30^\circ)^2$$

$$25 = 36 - 2 \cdot 6 \cdot b \cos 30^\circ + b^2 \cos^2 30^\circ + b^2 \sin^2 30^\circ$$

$$25 = 36 - 12b \cos 30^\circ + b^2$$

$$b^2 - 12b \cos 30^\circ = -11$$

$$b^2 - 12b \cdot \frac{\sqrt{3}}{2} + 11 = 0$$

$$b^2 - 6\sqrt{3}b + 11 = 0$$

$$|b| = 0.19615 \text{ m/hr}$$

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a)

->

$$\frac{x - L.F.P_x}{H.B.P_x - L.F.P_x} = \frac{y - L.F.P_y}{H.B.P_y - L.F.P_y}$$

When,  $x$  is temp in one scale and  $y$  is temp in another scale.

L.F.P = low freezing point  
H.B.P = High Boiling Point

$$\frac{x - L.F.P_x}{H.B.P_x - L.F.P_x} = \frac{y - L.F.P_y}{H.B.P_y - L.F.P_y}$$

b)

Given, For  $Z$ ,

$$U.F.P_Z = 180^\circ Z$$

$$L.F.P_Z = -20^\circ Z$$

For Celsius,

$$U.F.P = 100^\circ C$$

$$L.F.P = 0^\circ C$$

$$\text{GASE I, } x = y$$

$$\frac{x - L.F.P_Z}{U.F.P_Z - L.F.P_Z} = \frac{x - L.F.P_x}{U.F.P_x - L.F.P_x}$$

$$\frac{x + 20}{180 + 20} = \frac{x - 0}{100 - 0}$$

$$100(x + 20) = x \times (200)$$

$$100x + 2000 = 200x$$

$$2000 = 200x - 100x$$

$$2000 = 100x$$

$$x = 20 \text{ At } 20^\circ C$$

2 reading and 1 reading will be same

Q5C-2:

$$x = ?$$

$$y = 40^\circ$$

$$\frac{x+20}{180+20} = \frac{210}{100}$$

$$x+20 = \frac{4}{18} \times 200$$

$$x = 80 - 20$$

$$= 60^\circ \text{ \#}$$

Q1

Soln,  $\Delta\theta = \theta_2 - \theta_1$

$$= 40 - 20$$

$$= 20^\circ\text{C}$$

(As temp is increased it will lose time)

We know,  $T_{\text{today}} = \frac{1}{2} \Delta\theta \times 86400$

$$= \frac{1}{2} \times 1.9 \times 10^{-5} \times 20 \times 86400$$

$$= 16.416 \text{ sec}$$

It will lose 16.416 sec per day at  $40^\circ\text{C}$

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a) <sup>of body</sup>

→ water equivalent refers to the amount of water needed to absorb the same amount of energy <sup>negt</sup> that a body has gained or lost,

$$W_{\text{equiv}} = m s dt$$

$$W \times s_w dt = m s dt$$

$$W = \frac{m s}{s_w}$$

$$= \frac{C}{s_w}$$

$$W = \frac{C}{1000}$$

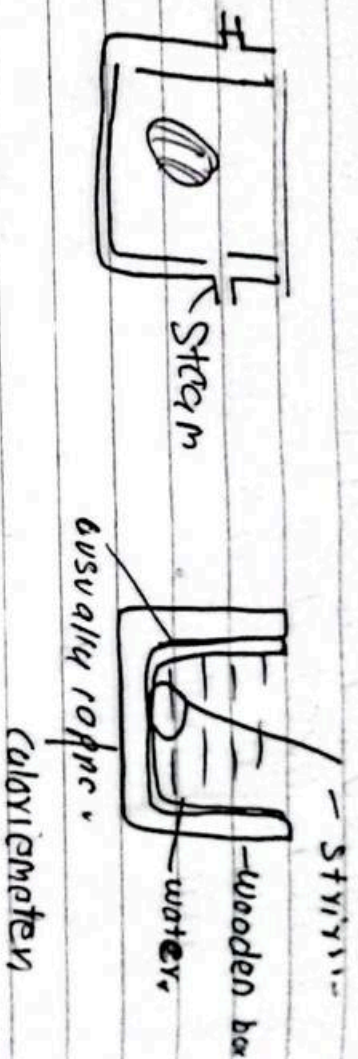
$$W = \frac{C}{1000} \quad \text{in S.I unit,}$$

→ It is different from heat capacity because heat capacity is the amount of heat needed to raise the temp of any mass of a substance by  $1^\circ\text{C}$ ,

$$C = m \cdot s$$

→ b)

## DETERMINATION OF S.P. HEAT CAPACITY OF A SOLID BY THE METHOD OF MIXTURES.



1) The substance whose S.P.C is to be measured is, heated up to certain temperature.

2) Then the substance is transferred to the calorimeter, and stirred until it reaches the final temp.

3) Then, the final temp of the mixture is noted.

Heat lost by substance = Heat gained by calorimeter and water.

Let, mass (mass of substance) =  $m_s$

(mass of calorimeter) =  $m_c$

(mass of water) =  $m_w$

S.P of substance =  $S_s$

S.P of calorimeter =  $S_c$

S.P of water =  $S_w$

Initial temp of substance  $\hat{=} \theta_s$

Initial temp of water and calorimeter  $\hat{=} \theta_{cw}$

Final temp of mixture  $= \theta$

Heat lost by substance = Heat gained by calorimeter and water,

$$m_s s_s (\theta_s - \theta) = m_c s_c (\theta - \theta_{cw}) + m_w s_w (\theta - \theta_{cw})$$

$$s_s (\theta_s - \theta) = \frac{m_c s_c (\theta - \theta_{cw}) + m_w s_w (\theta - \theta_{cw})}{m_s}$$

$$s_s = \frac{m_c s_c (\theta - \theta_{cw}) + m_w s_w (\theta - \theta_{cw})}{m_s (\theta_s - \theta)}$$

$$s_s = \frac{(\theta - \theta_{cw}) [m_c s_c + m_w s_w]}{m_s (\theta_s - \theta)}$$

C1

-> Given,

$m$

Mass of metal ( $m_m$ ) = 0.2 kg

Initial temp of metal ( $\theta_m$ ) = 100°C

Mass of water ( $m_w$ ) = 0.15 kg

Mass of

Water equivalent of calorimeter

$$(C_c) = 0.01 \text{ kg}$$

$$\text{or, } m_{csc} = 0.01$$

$$\text{final temp } (T) = 30^\circ\text{C}$$

$$S_w = 4200 \text{ J kg}^{-1}\text{K}^{-1}$$

Heat lost by metal = Heat gained by water + calorimeter

$$m_m S_m (\theta_m - \theta) = m_{csc} (\theta - 20) + m_{ws} (\theta - 20)$$

$$S_m = \frac{m_{csc} (\theta - 20) + m_{ws} (\theta - 20)}{m_m (\theta_m - \theta)}$$

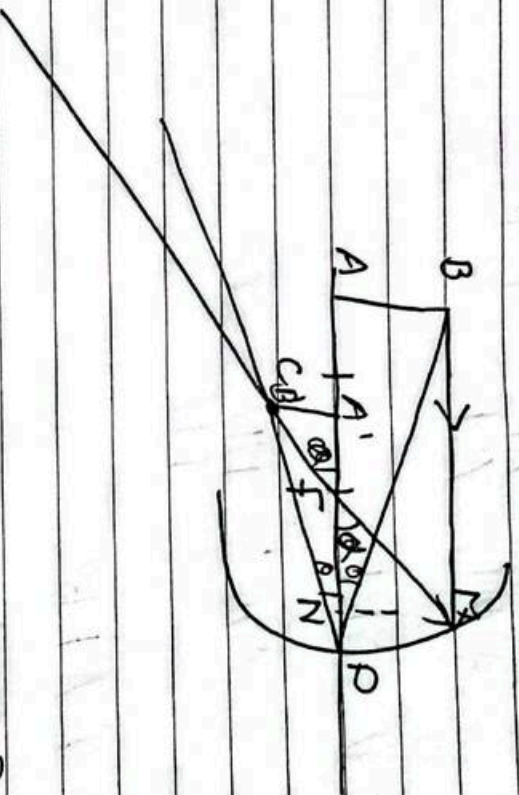
$$= \frac{0.01 (30 - 20) + 0.15 \times 4200 (30 - 20)}{0.2 (100 - 30)}$$

$$= 450.0071 \text{ J kg}^{-1}\text{K}^{-1}$$

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a1

Deriving mirror formula for a concave mirror forming a real image.



Let us consider a concave mirror of focal length  $f$ , center of curvature  $C$  and pole  $P$ , then, an object  $AB$  is placed behind  $C$ , then real and inverted image  $A'B'$  is formed between  $f$  and  $C$ ,

Let,  $u = AP$

$v = A'P$

$f = FP$

Now, In  $\triangle APB$

$$\tan \theta = \frac{AB}{AP} = (i)$$

In  $\triangle A'B'P$

$$\tan \theta = \frac{A'B'}{A'P}$$

From (i) and (ii)

$$\frac{AB}{AP} = \frac{A'B'}{A'P}$$

$$\frac{AB}{A'B'} = \frac{AP}{A'P} \quad \text{--- (iii)}$$

In  $\Delta MNE$ ,

$$\tan \alpha = \frac{MN}{FN} \quad \text{--- (iv)}$$

In  $\Delta A'EB'$ ,

$$\tan \alpha = \frac{A'B'}{A'F} \quad \text{--- (v)}$$

From, (iv) and (v)

$$\frac{MN}{FN} = \frac{A'B'}{A'F}$$

$$\frac{MN}{A'B'} = \frac{FN}{A'F} \quad \text{--- (vi)}$$

Since, the aperture is very small,

$$\left. \begin{aligned} FM &= FD \\ FN &= AB \\ FN &= FP \end{aligned} \right\} \text{--- (vii)}$$

Putting (vii) in eq (vi)

$$\frac{AB}{A'B'} = \frac{EP}{A'F} \quad \text{(viii)}$$

From,  $\phi$ -eq (viii) and (iii)

$$\frac{AB}{A'B'} = \frac{EP}{A'F}$$

$$\text{or, } \frac{u}{v} = \frac{f}{v-f}$$

$$uv - uf = vf$$

$$\frac{uv}{vf} = \frac{uf}{vf} = \frac{v}{v-f}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{v-f}$$

CONVENTIONS:

- i) The aperture is very small
- ii) The ray passes thru  $F$ ,
- iii) Real image is formed,
- iv) Obj is behind  $2F$ ,

b)

Given,

size of image = ( $S_i$ )

size of obj = ( $S_o$ )

$u = 12 \text{ cm}$

$f = ?$

Given,  $S_i = 4 S_o$

We know,

$$m = \frac{S_i}{S_o}$$

$$\frac{S_i}{S_o} = \frac{v}{u}$$

$$S_o = u$$

$$4 S_o = v$$

$$80 = v$$

$$v = 80$$

$$u$$

$$12 \times u = v$$

$$40 + 8 = v$$

$$v = 48 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{f} = \frac{1}{40} + \frac{1}{48}$$

$$= \frac{12 + 10}{480}$$

$$= \frac{22}{480}$$

$$\frac{1}{f} = \frac{11}{240}$$

$$f = \frac{240}{11}$$

$$f = 21.8 \text{ cm}$$

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a)

→ Coulomb's law states that, "the total force of attraction or repulsion is ~~div~~ between any two point charges is ~~div~~ proportional to the distance between them and the product of the two charges." directly proportional



From Coulomb's law,

$$F \propto \frac{K q_1 q_2}{r^2}$$

$$F = K \frac{q_1 q_2}{r^2}$$

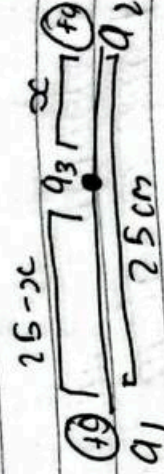
$$\frac{q^2 K}{r^2} = F \quad q^2 = \frac{F r^2}{K}$$

→ ~~to find~~

Coulomb's law of charge is defined as the charge of two point charges required to produce a force of attraction or repulsion that, puts them 1m apart, in vacuum or air.

b1

→ Given,



Given,

$$q_1 = 9 \times 10^{-6} \text{ C}$$

$$q_2 = 4 \times 10^{-6} \text{ C}$$

$$r = 25 \text{ cm},$$

$$x = ?$$

Force Electrostatic force produced betw  
een,  $q_1$  and  $q_3$   $(E_{13}) = K \frac{q_1 q_3}{(25-x)^2}$

Electrostatic force produced b  
 $q_3$  and  $q_2$ ;

$$E_{32} = K \frac{q_2 \cdot q_3}{x^2}$$

$$E_{13} = E_{32}$$

$$K \frac{q_1 q_3}{(25-x)^2} = K \frac{q_2 \cdot q_3}{x^2}$$

$$\frac{9 \times 10^{-6}}{(25-x)^2} = \frac{4 \times 10^{-6}}{x^2}$$

$$\left( \frac{x}{25-x} \right)^2 = \frac{4 \times 10^{-6}}{9 \times 10^{-6}}$$

$$\frac{x}{25-x} = \sqrt{\frac{4}{9}} \quad \frac{x}{25-x} = \frac{2}{3}$$

$$5x = 50$$

$$x = 10 \text{ m}$$

$$3x = 50 - 2x$$

~~The third charge~~ The point where the net force on 3<sup>rd</sup> charge is 0 is ~~also~~ 10 m from (4u, c).

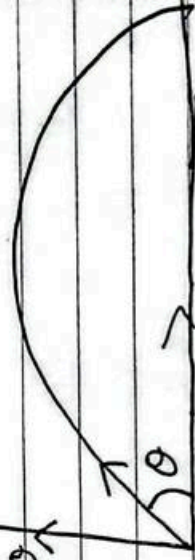
20)

a)

→ Suppose a body is projected from the ground with velocity  $u$  at an angle  $\theta$  with the horizontal, then  $u$  can be result into two rectangular components.

For ~~the~~ horizontal component,

~~the~~ velocity  $u_x = u \cos \theta$   $u_y = u \sin \theta$



For vertical,

$$u_y$$

$$x = u \cos \theta$$

velocity =  $u \sin \theta$  decreases

(Note: this velocity increases as it moves towards top and increases as it moves down.)

The eq of path of the projectile,

let  $p(x, y)$  be the position after time  $t$ .

The projectile travels,  $x$  ~~dis~~ horizontal distance and  $y$  vertical distance as a function of time.

For horizontal distance.

$$s_x = u_x t + \frac{1}{2} a_x t^2$$

$$s_x = x, \quad u_x = u \cos \theta, \quad a_x = 0$$

$$\begin{aligned} x &= u \cos \theta t \\ t &= x \quad - (1) \\ u \cos \theta \end{aligned}$$

For Vertical distance,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$y = u \sin \theta t - \frac{1}{2} g t^2$$

$$= u \sin \theta \cdot x \quad - \frac{1}{2} g x^2$$

$$\frac{u \cos \theta}{2} \quad \frac{u^2 \sin^2 \theta}{4 \cos^2 \theta} x^2$$

$$y = \tan \theta x + \left( \frac{-1}{2u^2 \cos^2 \theta} \right) x^2$$

The path of the projectile, is a parabola as, it takes the form of,

$$y = Ax + Bx^2 + C$$

Time of flight:

The total time it takes for the projectile to reach the top height and come down to the ground is called time of flight.

Since,  $v = 0 \text{ m/s}$ , when the projectile reaches the top most height,

$$v = u + at$$

$$0 = u \sin \theta - g t$$

$$t = \frac{u \sin \theta}{g}$$

Since the path of the projectile is symmetrical, the amount of time taken to reach top = amount of time taken to reach ground.

$$T = 2t \\ = \frac{2u \sin \theta}{g}$$

MAX height,

When the projectile reaches max height,  $v = 0 \text{ m/s}$ ,

$$v^2 = u^2 + 2as$$

$$0 = u^2 \sin^2 \theta - 2gH$$

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

## HORIZONTAL RANGE:

~~The horizontal~~ horizontal range takes total time ( $T$ ) to reach at the time of flight so,

$$\begin{aligned} R &= u \sin \theta \cdot T \\ &= u \sin \theta \cdot 2u \sin \theta \\ &= \frac{u^2 \sin 2\theta}{g} \cdot H \end{aligned}$$

b)

→ let us suppose,  $x$  be the range to be covered,

then,  $\theta_1$  be the 1<sup>st</sup> angle and  $\theta_2$  be the 2<sup>nd</sup> angle,

$$\begin{aligned} \text{let, } x &= 50\text{m} & t &= 10\text{sec} \\ \theta_1 &= 30^\circ \\ \theta_2 &= 60^\circ \end{aligned}$$

$$u_{01} = ?$$

$$u_{02} = ?$$

$$\text{Now, } x = \cancel{u \sin \theta} \frac{u \cos \theta}{t}$$

$$u_{01} = x t$$

$$\cos(30^\circ)$$

$$= \frac{50 \cdot 10}{\cos 30} = 577.35\text{ms}$$

$$u_{02} = \frac{50 \times 10}{\cos(60^\circ)} = 1000 \text{ m/s}$$

Now, when  $\theta = 30^\circ$

$$\begin{aligned} x &= u_0 \cos \theta \\ &= \frac{577.35 \times \cos(30^\circ)}{10} \\ &= 50 \text{ m} \end{aligned}$$

when,  $\theta = 60^\circ$

$$\begin{aligned} x &= \frac{u_0 \cos \theta}{t} \\ &= \frac{1000 \times \cos(60^\circ)}{10} \\ &= 50 \text{ m} \end{aligned}$$

$\therefore$  Hence, two different angles can obtain some horizontal range.

c)

→ Given,  $u_0 = 25 \text{ m/s}$   
 $\theta = 53^\circ$

i) Time of flight

$$T = \frac{2u_0 \sin \theta}{g}$$

$$= \frac{2 \times 25 \times \sin(53^\circ)}{10}$$

$$= 3.99 \text{ sec}$$

ii) MAX height:

$$H = \frac{u_0^2 \sin^2 \theta}{2g}$$

$$= \frac{25^2 \times (\sin 53^\circ)^2}{2 \times 10}$$

$$= 19.93 \text{ m}$$

iii) Max range,

$$R_{\text{max}} = \frac{u_0^2 \sin 2\theta}{g}$$

$$= \frac{25^2 \times \sin(2 \times 53^\circ)}{10}$$

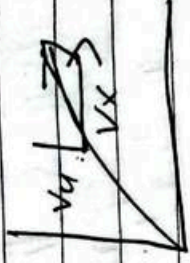
$$= \frac{25^2 \times 2 \sin 53^\circ \cos 53^\circ}{10}$$

$$= \frac{25^2 \times 2 \times 0.8 \times 0.6}{10}$$

$$= 60 \text{ m}$$

V after 1 sec,

$$t = 1 \text{ sec}$$



$$V = \sqrt{V_x^2 + V_y^2}$$

$$V_x = u \cos \theta$$

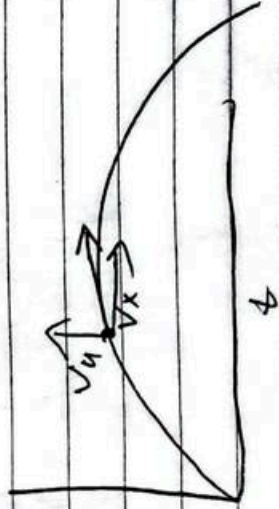
$$V_y = ?$$

$$V_y = u + at$$

$$V_y = u \sin \theta - gt$$

$$\begin{aligned} V &= \sqrt{(u \cos \theta)^2 + (u \sin \theta - gt)^2} \\ &= \sqrt{(25 \times \cos(57))^2 + (25 \times \sin(57) - 10)^2} \\ &= 17.48 \text{ m/s,} \end{aligned}$$

2)



2/1

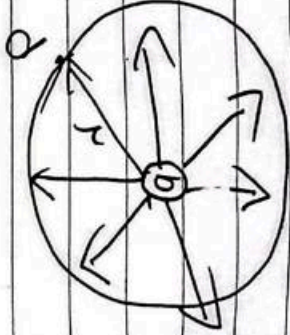
a)

→ Gauss theorem states that, "In any enclosed space the electric flux passing through that space in vacuum is  $\frac{1}{\epsilon_0}$  times the total charge enclosed by that space."

$$\Phi = \frac{q}{\epsilon_0}$$

Proving  $\Phi = \frac{q}{\epsilon_0}$

Let, ~~P be the~~ point charge of radius  $r$ , and,  $P$  be the point at the distance  $R$ , in which a gaussian sphere is drawn so that it encloses the  $q$ .



We know,  $\Phi = \sum A \cos \theta$

$$\Phi = \sum A \quad [\theta = 0 \text{ as, } 1] \\ = \sum 4\pi r^2$$

We know,

$$E = \frac{F}{q}$$

$$E = \frac{q}{4\pi r^2}$$

$$\Phi = \frac{q}{4\pi r^2}$$

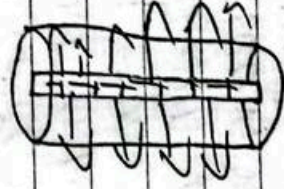
$$\Phi = \frac{q}{\epsilon_0} \quad \text{Proved \#}$$

b)

Let, the length of infinitely long straight uniformly charged wire, be  $L$

the, Surface charge density,

$$\sigma = \frac{q}{L}$$



Now, let us draw a cylindrical gaussian surface at a distance  $r$  from the rod, enclosing it,

$$\text{the, } \Phi_{\text{curved}} = \epsilon A$$

$$= \epsilon \times 2\pi r L$$

$$\Phi_{\text{cross section}} = \epsilon A \cos(90^\circ)$$

$$= 0$$

from,

$$E = \frac{q}{4\pi r^2 \epsilon_0} \quad \Phi = \frac{q}{\epsilon_0}$$

$$E = \frac{q}{4\pi r^2 \epsilon_0}$$

$$\Phi = \frac{q}{\epsilon_0}$$

$$E = \frac{q}{4\pi r^2 \epsilon_0}$$

$$\Phi = \frac{q}{\epsilon_0}$$

(1)  $\rightarrow$

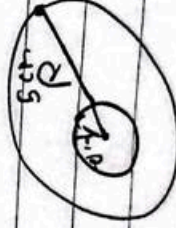
Two electric field lines never intersect each other because, the intersection of two electric field lines tangent drawn at the intersection gives us two direction, which ~~violates~~ violates vector ~~law~~ law of vector.

Q1  $\rightarrow$

Given,  $r = 0.1 \text{ m}$   
 $q = 2 \times 10^{-8} \text{ C}$

i)  $E$  at, 5 cm from it's center,

$$E = \frac{q}{4\pi r^2 \epsilon_0}$$



Here  $r = 0.1 + 0.05 = 0.15 \text{ m}$

$$\begin{aligned}\xi &= \frac{2 \times 10^{-8}}{4\pi r^2 \epsilon_0} \\ &= \frac{2 \times 10^{-8} \times 2 \times 10^{-8}}{(0.15)^2} \\ &= 8000 \text{ N/C}\end{aligned}$$

ii)  $\xi$  at 20 cm from its center.

$$\begin{aligned}r &= 0.1 + 0.2 \text{ m} \\ &= 0.3 \text{ m}\end{aligned}$$

$$\begin{aligned}\xi &= \frac{2 \times 10^{-8} \times 2 \times 10^{-8}}{(0.3)^2} \\ &= 2000 \text{ N/C}\end{aligned}$$

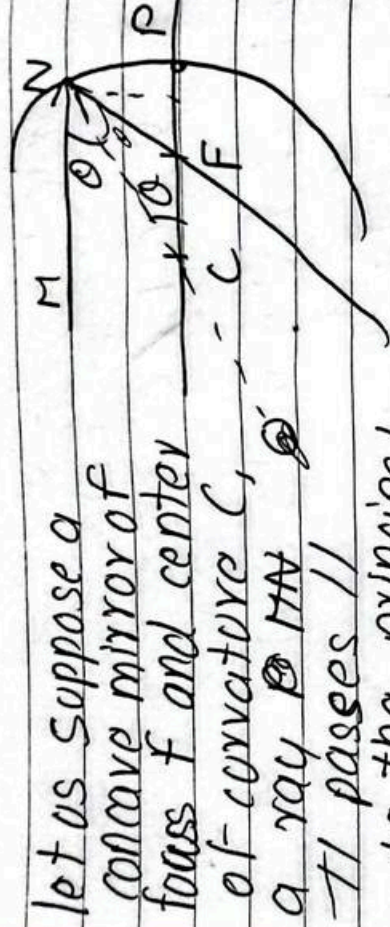
22)

9)

Radius of curvature is defined as the radius of the original sphere from which the spherical mirror was derived.

Focus length of a spherical mirror is defined as the length of focus from pole of the spherical mirror.

## R between C and f



let us suppose a  
concave mirror of  
focal  $f$  and center  
of curvature  $C$ ,  
a ray  $MA$   
// passes //  
to the principal  
axis, the passing  
thru,  $F$

let,  $R = CP$   
 $F = FP$

Now, ~~in  $\triangle MNO$~~

$\angle MNO = \angle ONF$  (i)  $\angle i = \angle r$   
 $\angle MNC = \angle NCF$  (ii) (Alternate)

from, (i) and (ii)

$\angle ONF = \angle NCF$ ,

as, two angles are equal  $\triangle NCF$  is  
an isosceles triangle,

$\therefore CF = FN$

Using convention, aperture is very  
small,

$$FN = FP$$

$$CF = FP$$

$$\text{No. } R = CP$$

$$= CF + FP$$

$$= FP + FP$$

$$= f + f$$

$$R = 2f \quad \underline{\underline{H}}$$

(b)

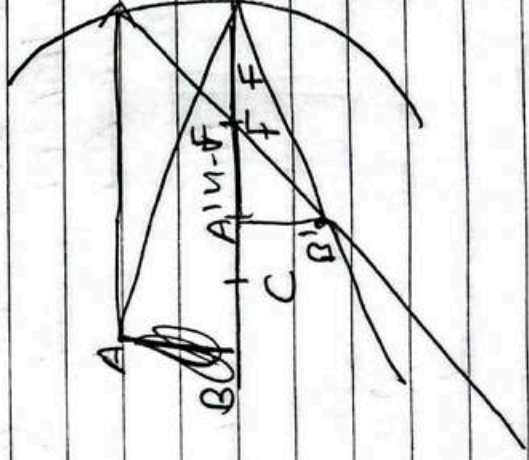
Now,

We know,

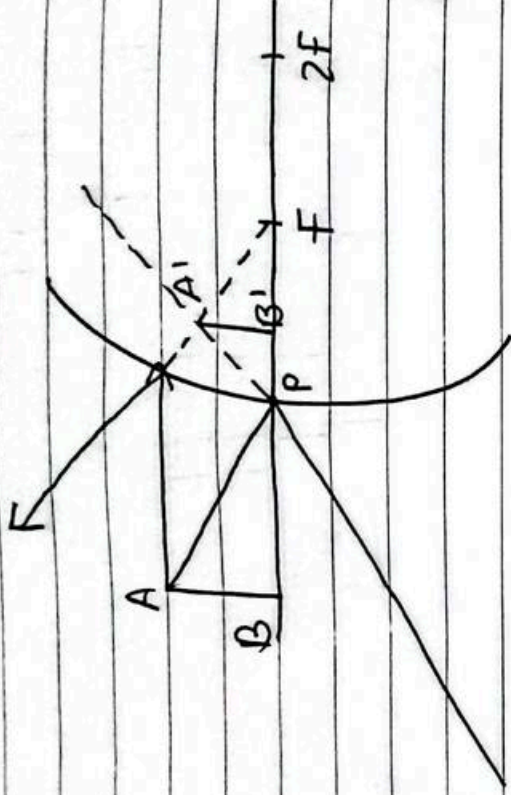
$$m = \frac{\text{image size}}{\text{obj size}}$$

$$= \frac{\text{image dis}}{\text{obj dis}}$$

$$m = \frac{V}{u}$$



$$\frac{f}{f} = \frac{1 + \frac{1}{V}}{\frac{1}{V}} \quad \therefore \quad \frac{f}{f} = \frac{f+1}{1}$$



Here, when, an image  $AB$  is placed in front of a convex mirror then, a image  $A'B'$  is formed in the other side.

Nature of image,

Size = diminished

Altitude = Erect

Location = diff side

Type = Virtual,

→ It is used as rear-view mirror because this mirror forms a diminished and virtual image which widens the view needed for driving to see most of the are a. ~~are~~

d)

→ Given,

img-obj height = 5cm  
image height = ?

$$u = 30\text{cm}$$

$$f = 20\text{cm}$$

We know,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= -\frac{1}{20} - \frac{1}{30}$$

$$\frac{1}{v} = \frac{30+20}{600} \quad v = -0.08333333$$

$$v = -12\text{cm}$$

$$v = -60$$

$v = -60\text{cm}$  (as convex mirror)

Now we know, magnification = image size  
obj size

$$x = \frac{v}{u}$$

$$x = \frac{-60}{30}$$

$$x = -2$$

$$x = -\frac{60}{30}$$

$$= -2$$

$$= 20\text{cm}$$

Hence, the image is diminished, erect,  
and virtual.